Drive-On-the-Line Project Report

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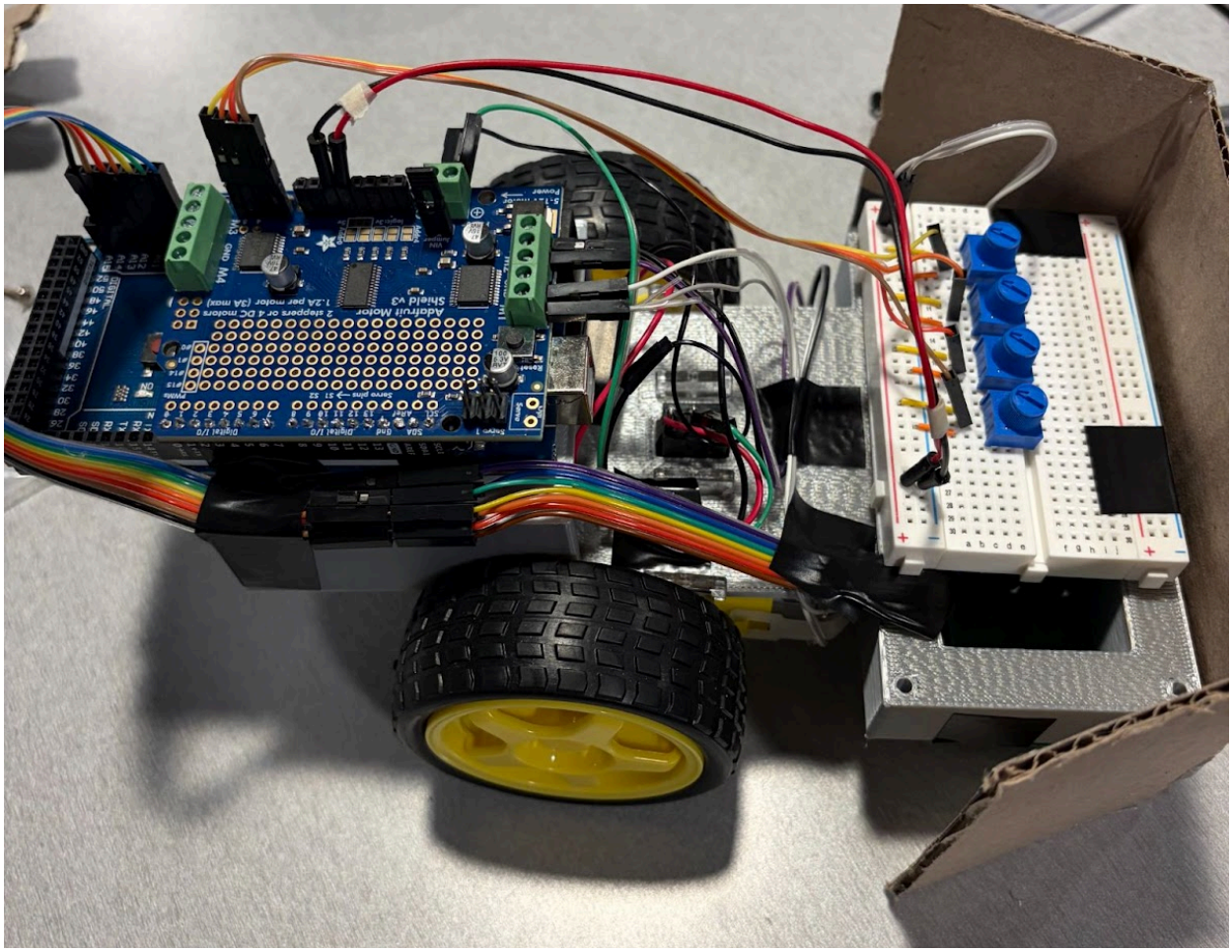

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EE 201: A Wi 26

INTRODUCTION

This project focused on designing and building a small robot that can follow a black track rink. The robot uses a combination of LEDs and photoresistors to detect the difference between the black line and the white surrounding surface, allowing it to adjust its direction and stay on the path. The overall goal of this project was to apply the circuit design concepts learned in EE 201 to a real hardware system.

To build the robot, we first designed and tested the sensing and motor control circuits, then assembled the hardware and integrated it with the robot chassis. After several rounds of testing and adjustments, the final robot was able to follow the black track with stable movement. While we didn't have the fastest robot, it was one of the smoothest and was able to complete the full track.

PROCEDURE

The robot was built through several stages, including sensor setup, pcb and chassis design, motor integration, and system calibration.

First, potentiometers were connected to the Arduino Mega using a breadboard. These potentiometers acted as variable resistors and allowed real-time adjustment of parameters such as speed and PID coefficients. The Arduino code was used to read the potentiometer values through the analog inputs and display them in a serial monitor to ensure they had accurate readings.

Next, we designed and soldered a PCB to support the photoresistors used to detect the black track. Each photoresistor was paired with a resistor to form a voltage divider circuit, and the outputs were connected to the Arduino analog pins. LEDs were also used, with about one per photoresistor, to illuminate the ground and get a better read on what is a dark surface and what is light. Before proceeding, the sensors were tested on different surfaces to test their ability to clearly distinguish between black and white surfaces.

Then, we soldered the motor shield and attached it to the Arduino Mega. The DC motors were connected to the motor shield outputs, and the Adafruit motor shield library was installed in the Arduino IDE to enable motor control through software.

After testing all our components, we designed a custom chassis for the robot using CAD software and 3D printed the structure. The printed chassis was then assembled and used as the main frame of the robot. The custom chassis featured a box-like structure above the back wheel where the 9V battery and 6V battery pack sources are stored, and the Arduino is elevated upon. On the front end, our robot had a rectangular extension that extruded towards the floor. On this, we affixed our custom PCB to the bottom, facing the floor, to keep the LEDs and photoresistors as close to the surface as possible. On the upper side of this extrusion, in line with the surface of the chassis, we affixed our breadboard potentiometer circuit. In addition to our motors, wheels, switch, and wiring, we also created and attached a double-layered and black-lined light shield to the front of our robot with the purpose of limiting shadows and interference from other light sources. To help counterbalance our robot since it was too forward heavy, we also attached two 9V batteries to the back of our chassis for weight and stabilization.

Finally, all code components were integrated onto the robot, and the sensors were calibrated using a lot of trial and error to get the robot to move smoothly. A PID control algorithm was implemented to calculate the error between the robot's position and the desired line. In the calibration phase, the motor speeds were adjusted based on this error so the robot could continuously correct its direction and follow the black track.

RESULTS

Challenge 1:

1. Twisting the potentiometers clockwise resulted in a decrease in values, while a counterclockwise twist resulted in an increase.
2. Switching the polarity of the potentiometers (changing VCC to GND) would provide opposite results when twisting the potentiometers.

Challenge 2:

1. The best distance for the sensors to properly differentiate between black and white surfaces is within about 8 cm from the surface.
2. Surrounding light and shadows affect the readings significantly. Strong light decreases the sensor readings, while shadows increase the readings.
3. The sensors are quite sensitive and are able to detect small changes in light intensity.

4. If the sensors are not reading the difference in colors properly, the readings can be improved by ensuring that the environmental brightness remains stable.
5. Adding an LED next to the photoresistors helps improve the readings because the LED provides a consistent light source and reduces the effect of changes in environmental brightness.
6. Sensor accuracy can also be improved by surrounding the sensors with paper or a small shield to further reduce the influence of the surrounding environment.

Challenge 3: ✓

Challenge 4: ✓

Challenge 5:

- All of our potentiometer knobs work. We are able to determine this by seeing their values increase, decrease and stay stable based on whether they are being twisted and in what direction on the serial monitor display.
- All of our photoresistors work and are able to sense the difference between a black and white surface. We are able to determine this by utilizing the serial monitor display and observing each one's associated analog voltage value in real time. This allows us to actively test each photoresistor in different states to confirm that a brighter environment results in a higher voltage and a darker one results in a lower voltage.
- We know each motor is connected correctly and turns when it should because whenever our robot was given a turn command, it turned in the correct direction consistently for both left and right.
- At the end of challenge 5, our cart was driving forward and backward as needed.

Challenge 6:

- We implemented a light shield, a custom PCB, an extra 9 V battery, and used a custom chassis as hardware additions. The problems our hardware attempted to solve were interference from other light sources disrupting light/dark surface detection, space efficiency, and wire organization of the photoresistor circuit, insufficient power supply for the overall robot functionality, and improving overall robot structure and mobility, respectively.
- When implementing the hardware additions, our chassis needed extra counterbalance support since our weight wasn't distributed evenly. We did this using a zip tie and extra batteries as back end weights to keep our robot from leaning forward and resting on our PCB. Additionally, we added a second layer to

our light shield and lined the inside with black tape to improve its functionality in reducing light interference. In the end all our hardware implementations were successful in their given purpose, even if they were a little inefficient.

Challenge 7:

- After calibration, the photoresistors read between 80 and 90 when a black surface is placed in front of them. When a white surface is placed in front of them they read less than 15.

Challenge 8:

- The PID values we had checked off by our TA are 10.00, 0.00, and 8.00 respectively.

Challenge 9:

The robot worked well at low speed, but some errors occurred after running dozens of times due to the decrease of battery voltage.

If we redo the project, two major improvements would be:

1. Improve the modelling of the car body by shortening and strengthening the structure for a better weight balance so as to avoid components touching the ground and creating friction that hampers the movement.
2. Shorten the PCB design, so that the light shield could be “tightened”. This would help the sensing when facing the “short U-turns”, for the other parts of the parth would not be covered within the light shield and influence the reading.

CONCLUSION

In this project, we designed and built a line-following robot capable of detecting and following a black track. Through creating custom additions, assembling the hardware, integrating sensors, and implementing PID control, we were able to create a system that could adjust its movement and stay on the track. The final robot was able to follow the line with relatively stable performance though slow speed.

This project helped us better understand how sensors, motor control, and feedback systems work together in a real system. We also gained experience troubleshooting hardware and tuning control parameters such as the PID coefficients. All of these are hands-on skills that are important for future work in electrical engineering, robotics, and embedded systems, particularly the ability to troubleshoot and adapt to unexpected

challenges on a team project.

REFERENCES

None :)